National University of Computer and Emerging Sciences
Lahore Campus 0092

Engineering Economics (MG3008)

Sessional-I Exam

Date: February 27 2025

Course Instructor(s)

Mr. Syed Haris Mujtaba Kazmi

Total Time (Hrs):	**1**
Total Marks:	**55**
Total Questions:	**3**

22L-6234 Roll No

6C Section

Student Signature

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Attempt all the questions.

CLO # 1: Recognize the fundamental economic and financial considerations involved in engineering projects

Q1: Describe the following.(Not more than few lines) [marks 15]

1) Opportunity Cost
2) Elasticity
3) Micro & Macro Economics

CLO # 2: Interpret the concept of time value of money

Q2: A Company's transactions for the last year are given below. [marks 20]

Year 1 through 5: Invest \$7000 in year 1, decreasing by \$1000 per year through year 5.

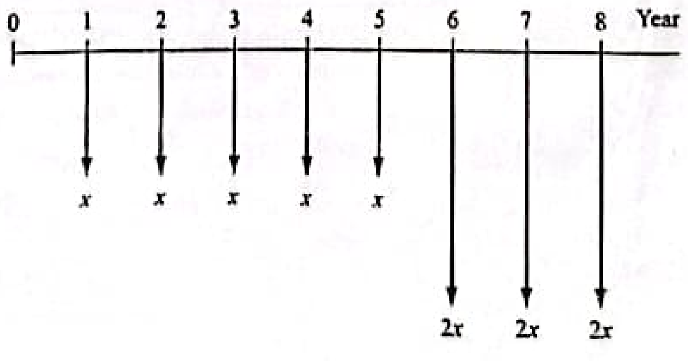
Year 6 through 10: No new investment and no withdrawals.

Year 11 through 15: Withdraw \$20,000 in year 11, decreasing 20% per year through year 15.

By using the time value of money, Determine the total present worth if the interest rate was 10 % per year.

CLO # 2: Interpret the concept of time value of money

Q3: For the cash flows shown in the diagram, Compute the value of x that will make the future worth in year 8 equals to \$-70,000. Use interest rate of 12% per year. [marks 20]



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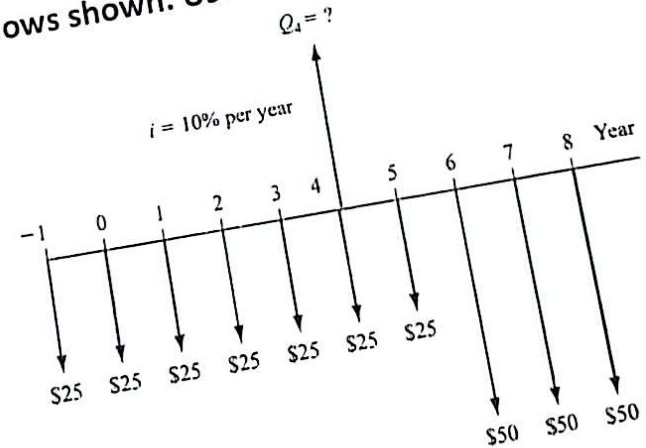
CLO #: 1

[20 marks]

Q1: Use the cash flow diagram to determine the single amount of money Q_4 in year 4 that is equivalent to all of the cash flows shown. Use i = 10 % per year.



CLO #: 1

[20 mar

Q2: A company budgeted $75,000 per year to pay for certain components over the next 5 years. If the company expects to spend $15,000 in year 1, how much of a uniform (arithmetic) increase each year is the company expecting in the cost of this part. Assume the Company uses an intere rate of 10% per year.

Attempt all the questions.

CLO # 2: Interpret the concept of time value of money

Q1: Maintenance Cost of a machine is projected to be \$100,000 in year 1 and increases by 4% each year through 5 years. What is the Future worth of the maintenance cost at an interest rate of 10% per year, compounded quarterly? [20 marks]

CLO # 2: Interpret the concept of time value of money

Q2: Find the capitalized cost of a present cost of \$300,000, annual costs of \$35,000, and periodic costs every 5 years of \$75,000. Use an interest rate of 12 % per year. [20 marks]

CLO # 3: Analyze cost effectiveness of alternatives using engineering economy methods

Q3: Determine which of the given mutually exclusive projects should be selected on the basis of incremental B/C ratio. [20 marks]

	Good	Better	Best	Best of All
FW of first cost, \$	10,000	8,000	10,000	14,000
FW of benefits, \$	19,000	13,000	27000	42,000
FW of disbenefits, \$	6,000	1,000	10,000	32,000
FW of M&O cost,\$	1,500	2,000	6,000	3,000